

Change of basis (Artin sections 3.5, 4.2)

Step 1 V v-space of dim n

old basis $B = (v_1, \dots, v_n)$

new basis $B' = (w_1, \dots, w_n)$

(Artin's terminology: hypervector
= list of vectors in a v-space
written in a row)

• Change of basis matrix

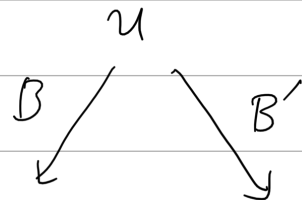
$$w_j = p_{1j}v_1 + p_{2j}v_2 + \dots + p_{nj}v_n = [v_1, v_2, \dots, v_n] \begin{bmatrix} p_{1j} \\ \vdots \\ p_{nj} \end{bmatrix}$$

$$[w_1 \dots w_n] = [v_1 \dots v_n] [p_{ij}] \quad \text{i.e.}$$

new basis = (old basis) (change of basis matrix) i.e.

$$B' = B P \rightarrow \text{equation is in hypervector notation} \\ (B, B' \text{ are hypervectors})$$

• old vs new coordinate



$$u = B \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = B' \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} \quad \left| \quad \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \xrightarrow{?} \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \bar{p}^{-1} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$
$$= B P \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} \quad \left| \quad u = \sum a_i v_i = [v_1 \dots v_n] \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

As B is a basis

$$B \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = B P \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = P \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

explanation

new coord = $\bar{p}^{-1}$ old coord.

old coordinates = (change of basis matrix) (new coordinates)

Step 2 Change of basis for a lin map $V \xrightarrow{T} W$

Question

How are matrices

$M_{B,C}(T)$ & $M_{B',C'}(T)$

related?

dim	p	q
old bases	B	C
new bases	B'	C'
change of basis matrices	P	Q

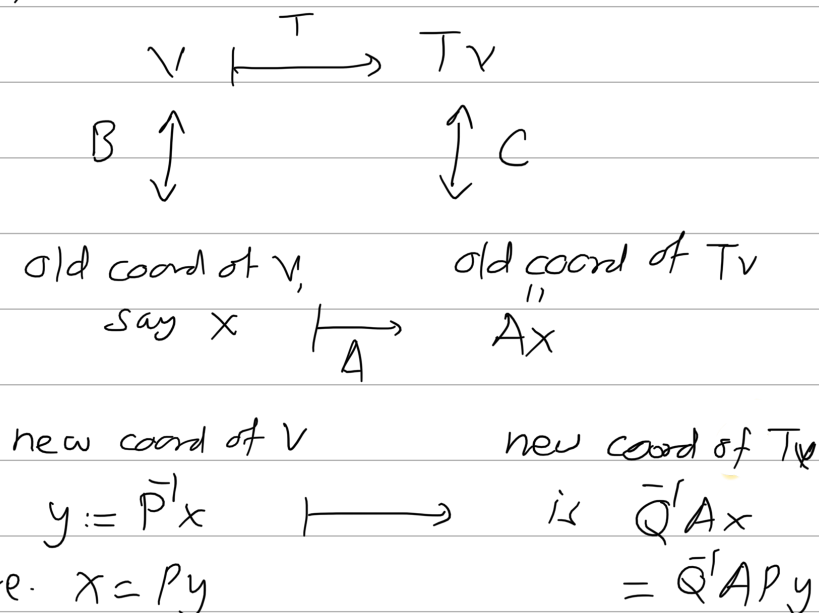
For simplicity $A = M_{B,C}(T)$

Answer: new matrix for T is $\bar{Q}'AP$

Fix this in your mind

One way: Take $V=W=\mathbb{R}$

old basis = 1



Spl. case $V=W$ i.e. T is an operator on V .

$$V \xrightarrow{T} V$$

IF old matrix of T (i.e. wrt B)

old basis B same B

is A , then new matrix (i.e.

new basis B' same B'

wrt B') is $\bar{P}'AP$

We saw several geometric examples of operators and their matrices wrt std basis. Try changing basis

Back to direct sum (3.6)

We saw external direct sum $W_1 \oplus W_2$ of any 2 v-spaces

If W_1, W_2 subspaces of same V , then we have a lin map

$$\begin{aligned} W_1 \oplus W_2 &\xrightarrow{S} V & \text{Im}(S) &= W_1 + W_2 \\ (w_1, w_2) &\longmapsto w_1 + w_2 \end{aligned}$$

Using this with dim formula we saw

$$\dim(W_1) + \dim(W_2) = \dim(W_1 + W_2) + \dim(W_1 \cap W_2)$$

Artin's pf. Take basis of $W_1 \cap W_2$ $\alpha_1, \alpha_2, \dots, \alpha_k$
Complete to a basis of W_1 $\beta_1, \dots, \beta_l$
Complete to a basis of W_2 $\gamma_1, \dots, \gamma_m$

Above eqⁿ becomes $(k+l) + (k+m) = \dim(W_1 + W_2) + k$
ie. formula is eqv to $\dim(W_1 + W_2) = k + l + m$

To prove this formula, show that α 's β 's γ 's form a basis of $W_1 + W_2$

Internal direct sum: V is called internal direct sum of its subspaces $W_1, W_2, \dots, W_k$ if $\begin{cases} W_i \text{ are lin indep} \\ \text{and span } V. \end{cases}$

• Observe: In that case ① $\dim V = \sum_i \dim W_i$ b/c
U basis of W_i is a basis of V

② In general I have a linear map $W_1 \oplus \dots \oplus W_k \xrightarrow{S} W_1 + \dots + W_k$
 $(w_1, w_2, \dots, w_k) \mapsto \sum_i w_i$

1 S is surjective $\Leftrightarrow W_1, \dots, W_k$ span V

S is injective $\Leftrightarrow$ ——— are lin indep.

Thus $V =$ internal direct sum of subspaces $W_1, W_2, \dots, W_k$

$\Rightarrow V \cong W_1 \oplus \dots \oplus W_k$ via the sum map

Sp1 case $k=2$ W_1, W_2 lin indep $\Leftrightarrow$ check $W_1 \cap W_2 = 0$

W_1, W_2, W_3 lin indep $\Rightarrow W_1 \cap W_2 = 0, W_2 \cap W_3 = 0$
 $W_1 \cap W_3 = 0$

BUT converse not true! Example?